

LAND RESOURCES

Land is a major constituent of life supporting system – the lithosphere. It forms about 1/5th of the earth surface, which is largely covered with natural forests, grasslands, wetlands, agricultural land and urban and rural settlements.

The fertile surface layer of earth capable of supporting plant life is called soil. Plants and animal materials decay are released into the nutrient bank in the soil.

Land Degradation: A number of factors are responsible for the degradation of land.

These include

- Soil erosion,
- Water logging,
- Salination,
- Shifting cultivation,
- Desertification and
- Various developmental activities.

Soil Erosion – Removal of top fertile layer of the soil by water, wind oceanic waves and glaciers is called soil erosion.

- Erosion of soil by water generally occur near hills where high speed streams and flooding removes top soil.
- Strong winds also erode the soil and bring sand from deserts to adjacent fertile land, converting the latter into desert (Example Thar Desert in Rajasthan).
- Various human activities such as felling of trees, over-grazing, over cropping and improper tilling accelerate soil erosion.

Desertification – Transformation of fertile land into a desert by natural or man's activities is called desertification. It can result from various causes such as erosion of top soil, shifting of sand dunes by wind and overgrazing in lands sparsely covered by grass. Many deserts of the world have developed by the human activities.

Soil Conservation

1. **Conservational Tillage** – The incorporation of residues from previous crops into the soil by ploughing is called conservational tillage. It improves soil permeability and increases the organic matter which in turn improve soil moisture and nutrients.
2. **Organic Farming** – Application of biofertilizers is an important practice of organic farming. The organic farming has long term beneficial effects on soil fertility.
3. **Crop Rotation** – It is a practice of growing different crops in successive years on the same piece of land. It decreases soil loss and prevents depletion of nutrients from the soil.
4. **Contour Ploughing** – Rain water is caught in circular and held in furrows and stored which reduces runoff and erosion.
5. **Contour Terracing** – A slope is divided into a number of small flat fields called terraces. The terraces slow down the velocity of the run off and allow the water to move to the sides of the fields where it flows away without eroding the soil.
6. **Afforestation** – Trees or windbreaks are planted in deserts which check the velocity of wind.

Landslides – Landslides is the moving down of rock of soil mass due to gravitational pull. Various anthropogenic activities like hydroelectric project, large dams, reservoirs, construction of roads and railway lines, mining etc. make mountain slopes brittle leading to landslides.